

Graded Containment with an Observability Price: Sequential Response to Ambiguous Trading Incidents

Chiheb Mejrissi^{a,*}

^aDepartment of Computer Science, Example University, 1 Example Street, Example City, Country

Abstract

An anomaly on a trading gateway may be a compromise or a fault; the usual response is a kill-switch. We model the response as a partially observed Markov decision process over the hidden cause, with graded actions and optionally a benign market shock as third cause. Throttling reduces harm but blinds: it scales order-flow separation by a discriminability factor $\kappa \in [0, 1]$. Classical results give a critical factor κ^* below which an optimal operator need never throttle, and recovers our companion paper's one-shot verification rule when order flow is uninformative. New to graded containment, $\kappa^* > 0$ whenever throttling has a disruption cost; in a diffusion version with a disruption cost or residual harm, throttling pays exactly when its running cost per unit of information is below monitoring's, giving κ^* in closed form. An exact computation with prespecified criteria, on the companion study's cells where the classifier has no supported improvement over the best constant policy, supports three claims: the graded policy materially beats the best Bayesian kill-switch at a share 0.796 of converged points, with little gain from the throttle; $\kappa^* < 1$ on a small part of this

*Corresponding author.

Email address: chiheb.mejrissi@example.org (Chiheb Mejrissi)

region; there, ignoring the observability price loses to the kill-switch. The gain over the one-shot rule is material at only a share 0.183 (weakened); a benign third cause’s effect is inconclusive at this power; the claim that the diffusion law locates the discrete κ^* is refuted, though a closed-form policy from it recovers most of the graded gain.

Keywords: OR in security, partially observed Markov decision process, graded containment, incident response, electronic trading

1. Introduction

An alarm on an electronic trading gateway rarely comes with its cause attached. The same burst of abnormal orders can be the work of an adversary acting through a legitimate session (a *compromise*) or the symptom of a benign malfunction (a *fault*). A compromise should be contained quickly, while containing a fault disrupts a legitimate client for nothing. The operator’s standard tool is a kill-switch, which stops the flow once the evidence crosses a threshold. Between doing nothing and killing, an operator can also throttle the flow, which limits harm, or check the session’s credentials on a channel that order flow does not carry.

This paper asks when such a graded response is worth more than the best kill-switch. The answer turns on a feature that a static rule cannot see: containment changes what the operator observes next. Paper 1 (Mejrissi, 2026b) shows that two causes that give the same immediate output can still be told apart through the feedback of the incident’s injection into the latent state of the book (its Theorems 2 and 3). A throttle can remove that evidence. We measure what it keeps by the *discriminability factor* κ , the share of the per-period order-flow separation between the causes that a throttled period keeps (Definition 3). In Paper 1’s linear-Gaussian model,

$\kappa \leq 1$ (Assumption 4) follows for a throttle that scales the injection and, for pairs confounded at the output level, for one that suppresses feedback from the onset; otherwise it is an assumption of this paper’s model (Section 3.7).

We model the response as a finite-horizon partially observed Markov decision process (POMDP) over the hidden cause (Section 3), with five actions: **monitor**, **throttle**, **verify** (the costed, delayed check of Paper 2 (Mejrissi, 2026c)), **kill** and **leave**. It is solved exactly by backward induction on a belief grid. A benign market shock can be added as a third cause; the belief then lives on the 2-simplex.

Contributions.

- *Model and structure* (Sections 3–4). Containment scales the harm rate by a harm factor ρ and the order-flow separation by κ . Classical results, specialised and not claimed as new, make the critical factor κ^* a single threshold below which an optimal operator need never throttle, and embed Paper 2’s one-shot rule exactly. Specific to graded containment, a throttle that removes all evidence is never strictly useful, and with a disruption cost $\kappa^* > 0$ (Proposition 6).
- *A cost-per-information law* (Proposition 5): in a diffusion version of the two-state model with a disruption cost or residual harm, the throttle is worth using at a belief exactly when its running cost per unit of information is below monitoring’s. This gives κ^* in closed form, shows that a scaling throttle is worth using only if harm falls faster than quadratically in the flow, and that an operator who plans as if throttling kept full discriminability has a cost that grows without bound as κ falls.

- *A numerical test fixed in advance* (Section 5), on a region calibrated from the set $\mathcal{F}$ of Paper 1’s cells without a supported improvement over the best constant policy. The graded policy beats the best Bayesian kill-switch strictly and materially at most points, with little of the gain from the throttle; $\kappa^* < 1$, and the blind operator does worse than the kill-switch, on a non-empty but small part of the region; the gain over Paper 2’s one-shot rule is weakened, and the effect of a benign third cause is inconclusive at this power. That the law of Proposition 5 locates the discrete κ^* is refuted, while a closed-form policy derived from it recovers most of the graded gain.

The comparison is against the *best* Bayesian kill-switch, an optimal stopping rule on the belief; weak dominance holds by construction, and what is claimed is *strict* dominance on a stated region.

2. Related work

POMDP intrusion response. The closest precedent treats intrusion prevention as an optimal stopping problem (Hammar and Stadler, 2022): a defender with several progressively stronger stops faces a partially observed intrusion, and the optimal policies are thresholds in the belief. There the hypotheses are intrusion against no intrusion, there is no verification on a separate channel, and a defensive action does not degrade what the defender observes next; here the causes are two anomalies whose origin is in question, and throttling multiplies the order-flow separation by κ . Miehling et al. (2018) model network defence on an attack graph as a POMDP that trades security against availability, and defence and the allocation of limited monitoring are chosen jointly over Bayesian attack graphs (Kazeminajafabadi

and Imani, 2024); monitoring there is a separate, budgeted resource, whereas here observability falls as a side effect of containment. Exactness is not itself the contribution: a causal POMDP model of the CAGE-2 benchmark already admits an online tree-search method with a provably optimal defender strategy (Hammar et al., 2026).

Verification, sequential testing and the cost of information. The verify action is the costed, delayed check of Paper 2 (Mejrissi, 2026c), which chooses once between leaving, verifying and containing; Proposition 3 recovers that rule exactly when order flow is uninformative and $T \geq \ell_v$. In the sequential design of experiments and active hypothesis testing (Chernoff, 1959; Naghshvar and Javidi, 2013; Nitinawarat et al., 2013), actions buy information, the error exponent is governed by the best Chernoff information over controls, and skipping an observation saves its cost without reducing harm. Here the most informative posture, **monitor**, is also the most harmful under a compromise, and the action that reduces harm reduces information. Theorem 1 rests on Blackwell’s ordering (Blackwell, 1953) and on structural results for POMDPs and controlled sensing (Krishnamurthy, 2025). Proposition 5 uses the time change to the information clock of Campbell et al. (2025), who show, for a Brownian drift whose signal intensity is controlled at a running cost that does not depend on the belief, that the optimal control is the constant minimiser of the cost per unit of information rate; Harrison and Sunar (2015) study a binary unknown learnt through several modes of different cost and quality. In these the cost of an experiment does not depend on the belief and reduces no harm; in our model it does both.

The price of observability. A watermark can maximise the Kullback–Leibler divergence for a given increase in LQG cost (Naha et al., 2023), an explicit

price of observability; our throttle runs the other way: it buys harm reduction and pays in discriminability. Section S5 of the Supplementary Material gives the full positioning.

3. Model

An operator watches an anomaly on a trading gateway whose cause, a compromise K or a fault F (in the extension also a benign market shock B), it does not see. At each period it observes an order-flow statistic and chooses an action, until a terminal action ends the episode or the horizon is reached.

3.1. Time, causes and actions

Definition 1 (Time, hidden cause, model variants). Time is discrete, $t = 0, 1, \dots, T$, with horizon $T \geq 1$; decisions are taken at $t = 0, \dots, T - 1$ and the incident is under way from $t = 0$ (the alarm time). The hidden cause M is drawn once at $t = 0$ and does not change. In the *two-state model* $M \in \mathcal{M}_2 = \{K, F\}$ (compromise, fault) and the belief is the scalar $p_t = P(M = K \mid \mathcal{I}_t) \in [0, 1]$, where $\mathcal{I}_t$ is the history of actions and observations before the decision at t . In the *three-state model* $M \in \mathcal{M}_3 = \{K, F, B\}$, where B is a benign market shock, and the belief is the point $\pi_t = (\pi_t(M))_{M \in \mathcal{M}_3}$ of the 2-simplex $\mathcal{S}_2$. The prior is p_0 (resp. π_0).

Definition 2 (Actions). At each decision time the operator chooses one action from

$$\mathcal{A} = \{\text{monitor, throttle, verify, kill, leave}\}.$$

monitor and throttle are *postures* held for one period, and verifying is the posture held while verification runs; verify is the verification action of Paper 2

(Definition 2 there), available at most once per episode, which holds the *verifying* posture for $\ell_v \geq 1$ periods and returns its outcome at the end of them (no decision is taken during these periods); *kill* (contain) and *leave* (stand down) are *terminal* and end the episode. At $t = T$ only *kill* and *leave* are available. A *verification flag* $v_t \in \{0, 1\}$ records whether verification has been used; the decision state is (t, p_t, v_t) (resp. (t, π_t, v_t)).

throttle limits the flow, for instance by rate limits, at a disruption cost; *kill* suspends the session and *leave* stands down, which the embedding of Paper 2's rule needs (Proposition 3).

3.2. Observations and the discriminability factor

Definition 3 (Order-flow observations and separation). In each period spent in a posture $x \in \{\text{monitor}, \text{throttle}\}$ the operator observes an order-flow statistic $y \in \mathbb{R}$; in a period of the *verifying* posture it observes one with the law of the *monitor* posture if $\omega = 1$ and none if $\omega = 0$, for a fixed $\omega \in \{0, 1\}$. In the two-state model,

$$y \mid (M = K, x) \sim \mathcal{N}(\tfrac{1}{2}d_x, 1), \quad y \mid (M = F, x) \sim \mathcal{N}(-\tfrac{1}{2}d_x, 1), \quad d_{\text{monitor}} = d, \quad d_{\text{throttle}} = \kappa d,$$

with per-period separation $d \geq 0$ and *discriminability factor* $\kappa \geq 0$. In the three-state model, $y \in \mathbb{R}^k$ ($k \in \{1, 2\}$) with $y \mid (M, x) \sim \mathcal{N}(g_x \mu_M, I_k)$, $g_{\text{monitor}} = 1$, $g_{\text{throttle}} = \kappa$, for fixed means $\mu_K, \mu_F, \mu_B \in \mathbb{R}^k$; the pairwise per-period separations are $d_{MM'} = \|\mu_M - \mu_{M'}\|$ under monitoring and $\kappa d_{MM'}$ under throttling. The two-state model is the three-state model restricted to $\{K, F\}$ with $\|\mu_K - \mu_F\| = d$: for $d > 0$ the statistic $u^\top (y - g_x(\mu_K + \mu_F)/2)$, $u = (\mu_K - \mu_F)/d$, is sufficient for $\{K, F\}$ and has the two-state laws above.

In Paper 1's linear model κ is the ratio $\kappa_W = d_W(\Theta)/d_W(\emptyset)$ of whitened separations with and without the throttle (Definition 11, Supplementary Material).

Definition 4 (Verification outcome). The outcome $Z \in \{+, -\}$ of **verify** has the law of Paper 2, Definition 2: $P(Z = + \mid K) = s$, $P(Z = - \mid F) = t_F$. In the three-state model $P(Z = - \mid B) = t_B$, with $t_B = t_F$ unless stated otherwise. During the ℓ_v periods of the **verifying** posture, order flow is observed as in Definition 3 and harm accrues as in the **monitor** posture (Definition 6); $\omega = 0$ is the analogue of Paper 2, Assumption 4.

Definition 5 (Belief update). After an order-flow statistic y observed in posture x , the two-state belief is updated by Bayes' rule (in the **verifying** posture with $\omega = 1$, $d_x = d$),

$$\text{logit } p' = \text{logit } p + d_x y,$$

(the log-likelihood ratio of the two Gaussians of Definition 3), and after a verification outcome z by $p' = p^z$ with p^+, p^- as in Paper 2, Definition 2. In the three-state model, $\pi'(M) \propto \pi(M) \phi_k(y - g_x \mu_M)$ after y and $\pi'(M) \propto \pi(M) P(Z = z \mid M)$ after z , where ϕ_k is the standard normal density on $\mathbb{R}^k$.

3.3. Costs and optimal value

Definition 6 (Costs). The terminal costs form the loss table (C_K, C_F, H_K, H_F) , extended by (C_B, H_B) in the three-state model: **kill** (containing) at any t costs C_M under cause M and ends the episode; **leave** (standing down) at any t costs H_M and ends it. Write $a = H_K - C_K$ (what containing saves under compromise) and $b = C_F - H_F$ (what containing costs needlessly under a fault). Each period spent in a posture adds a stage cost: **monitor** costs r_M (the harm accrued in the period under cause M), **throttle** costs $\delta + \rho r_M$, with disruption cost $\delta \geq 0$ and *harm factor* $\rho \in [0, 1]$; **verify** costs c_v once plus r_M for each of its ℓ_v periods in the **verifying** posture. The baseline harm rates are $r_K \geq 0$ and $r_F = r_B = 0$. Stage costs are additional to the

terminal cost, whichever terminal action ends the episode, and there is no discounting.

The throttle trades a certain disruption cost δ against a reduction of expected harm from pr_K to ρpr_K , and pays for it with evidence, through κ ; the value of throttling of Definition 10 measures whether that trade is worth making.

Definition 7 (Policies, value, Bellman recursion). A policy maps each decision state to an action. Its *expected cost* $J^\pi(p_0)$ is the expected sum of stage and terminal costs from prior p_0 with $v_0 = 0$. In the two-state model the optimal value $V_t(p, v)$ satisfies $V_T(p, v) = R(p) = \min\{pC_K + (1 - p)C_F, pH_K + (1 - p)H_F\}$ and, for $t < T$,

$$V_t(p, v) = \min_{a \in \mathcal{A}_t(v)} Q_t(p, v, a),$$

$$Q_t(p, v, \text{kill}) = pC_K + (1 - p)C_F, \quad Q_t(p, v, \text{leave}) = pH_K + (1 - p)H_F,$$

$$Q_t(p, v, \text{monitor}) = pr_K + \mathbb{E}_y[V_{t+1}(p', v)],$$

$$Q_t(p, v, \text{throttle}) = \delta + \rho pr_K + \mathbb{E}_y[V_{t+1}(p', v)],$$

$$Q_t(p, 0, \text{verify}) = c_v + \ell_v pr_K + \mathbb{E}[V_{t+\ell_v}(p'', 1)],$$

where p' is the update of Definition 5 in the posture taken, the expectation is over y under the predictive mixture $p\mathcal{N}(\frac{1}{2}d_x, 1) + (1 - p)\mathcal{N}(-\frac{1}{2}d_x, 1)$, p'' is the belief after the verification outcome (and, if $\omega = 1$, the ℓ_v observations of the verifying posture), $\mathcal{A}_t(1)$ excludes `verify`, and `verify` is excluded when $t + \ell_v > T$. The three-state recursion is the same with p replaced by $\pi \in \mathcal{S}_2$ and every $pX_K + (1 - p)X_F$ by $\sum_M \pi(M)X_M$. Ties are broken towards the lower rung of the order `leave`, `monitor`, `verify`, `throttle`, `kill`.

3.4. Numerical solution

The recursion is solved by backward induction on a belief grid with linear interpolation; the grid, the quadrature rules and the criterion of grid convergence at tolerance ϵ are Definition 8 (Section S3 of the Supplementary Material). In the two-state model the quadrature rule is Bayes-consistent, so that the grid problem is itself an exact Bayes recursion and concavity of the grid values (Proposition 1(iii)) is a checkable property of the solver’s output. The solver is written in NumPy; the code, with the version used for every reported computation, is to be made available in an archived repository.

3.5. Baselines and reported quantities

Definition 9 (Baseline policies). (i) *Fixed-threshold kill-switch* KS_θ : **monitor** while $p_t < \theta$, **kill** as soon as $p_t \geq \theta$, and at $t = T$ the better terminal action at p_T . (ii) *Best Bayesian kill-switch* KS^* : the optimal policy of the recursion of Definition 7 with the action set restricted to $\{\text{monitor}, \text{kill}, \text{leave}\}$. Its value is V_t^{KS} ; by construction $V_0^{\text{KS}} \leq J^{\text{KS}_\theta}$ for every θ . (iii) *Kill-switch with verification* KSV^* and *kill-switch with throttle* KST^* : the same with **verify**, respectively **throttle**, added to the restricted set; their values are V_t^{KSV} and V_t^{KST} . (iv) *Paper 2’s one-shot rule* P2 : at $t = 0$, with belief p_0 , **verify** if p_0 lies in Paper 2’s verification region S (Paper 2, Definition 3, with Paper 2’s delay harm read as $\ell_v r_K$), then take the better terminal action at the posterior; otherwise take the better terminal action at p_0 . No order flow is used. (v) *Observability-blind throttle policy* NB : at each decision state (t, p_t, v_t) it takes the action of the optimal policy of Definition 7 computed with $\kappa = 1$ (tie-breaking of Definition 7), where p_t is the Bayes belief of the true model: in throttled periods it updates with the true separation κd

(Definition 5). Its expected cost J^{NB} is evaluated in the true model by the recursion of Definition 7 with the minimum replaced by NB's action.

KS* is the strongest kill-switch the model allows; KSV* and KST* add one rung each, which splits the gain of the full ladder into a verify and a throttle share; NB is a correct filter with a misspecified planner, whose cost is evaluated in the true model.

Definition 10 (Operating characteristics, gains, critical discriminability factor). For a policy π and prior p_0 : the *false-kill rate* $\text{FPR}^\pi = P^\pi(\text{kill} \mid F)$; the *missed-containment rate* $\text{FNR}^\pi = P^\pi(\text{no kill} \mid K)$; the *expected time to a terminal action*; and the *expected throttled time*. The *graded gain* is $\Gamma(p_0) = V_0^{\text{KS}}(p_0, 0) - V_0(p_0, 0) \geq 0$, with the decomposition into the verify share $V_0^{\text{KS}} - V_0^{\text{KSV}}$ and the throttle share $V_0^{\text{KS}} - V_0^{\text{KST}}$. The *value of throttling* at discriminability factor κ is $G(\kappa) = V_0^{\text{KSV}}(p_0, 0) - V_0(p_0, 0; \kappa) \geq 0$, where V^{KSV} (Definition 9(iii)) is the optimal value with every action but throttle (it does not depend on κ). The *critical discriminability factor* is $\kappa^* = \inf\{\kappa \in [0, 1] : G(\kappa) > 0\}$ ($\kappa^* = 1$ if the set is empty).

Every comparison is in expected cost at the prior. *Dominance* means strictly lower expected cost at the prior, $\Gamma(p_0) > 0$ for the graded gain, and is claimed only on a stated region of parameters.

3.6. Assumptions

Assumption 1 (Cost orderings). $H_K > C_K$ and $C_F > H_F$, that is $a > 0$ and $b > 0$ (Paper 2, Assumption 1); in the three-state model also $C_B > H_B$. In the two-state model the *kill/leave indifference belief* is $q^* = b/(a + b) \in (0, 1)$, the unique belief at which kill and leave have the same expected terminal cost: $(pH_K + (1 - p)H_F) - (pC_K + (1 - p)C_F) = (a + b)(p - q^*)$. Stage costs satisfy $r_K \geq 0$, $\delta \geq 0$, $c_v \geq 0$, $\rho \in [0, 1]$.

Assumption 2 (Conditionally i.i.d. Gaussian order flow). Given M and the postures taken, the per-period order-flow statistics are independent, with the laws of Definition 3; the separation d does not change with time.

Assumption 3 (Separate channel). The verification outcome Z is conditionally independent of all order-flow statistics given M (Paper 2, Assumption 2), and $s + t_F > 1$.

Assumption 4 (Throttle effects). $0 \leq \kappa \leq 1$: throttling never makes the causes easier to tell apart. ρ and κ are independent parameters.

Assumption 5 (Known model; non-strategic cause). All parameters ($d, \kappa, \rho, \delta, s, t_F, c_v, \ell_v$, the loss table, the prior) are known to the operator, and the cause does not react to the operator’s actions (a strategic attacker is outside this paper’s scope).

In Paper 1’s linear-Gaussian model the separation can grow with time through state feedback; a time-varying separation d_t is a stated variant of Assumption 2. Assumption 4 excludes throttles that make the causes easier to tell apart; Section 3.7 shows that it is a restriction of the model, not a consequence of Paper 1’s.

Remark 1 (Equivalent-Gaussian calibration). Paper 1’s ABIDES constructions were never shown to be Gaussian. A separation backed out of a Paper 1 accuracy through the Gaussian benchmark $\text{Acc}^* = \Phi(d_W/2)$ is an *equivalent-Gaussian calibration heuristic*, and it is a separation d_W for the classifier’s whole observation window, not a per-period separation. Under Assumption 2, n_W periods of monitoring give separation $d\sqrt{n_W}$, so the heuristic per-period value is $d = d_W/\sqrt{n_W}$, where n_W is the number of decision periods the window is taken to span (a modelling choice, stated wherever used). Every result evaluated at a calibrated d is also reported

over a direct sweep of d .

3.7. Where $\kappa < 1$ comes from

In the linear-Gaussian model of Paper 1 (Mejrissi, 2026b), Definition 11 and Proposition 4 (Section S1 of the Supplementary Material) model two throttles. A throttle that scales the injection by r has $\kappa = r$, so its harm and discriminability factors move together. A throttle that removes the injection’s feedback into the state, applied from the onset to a pair confounded at the output level, has $\kappa = 0$: all the discriminability of monitoring then comes from that feedback, and the condition under which monitoring separates such a pair is the negation of the condition of Paper 1’s Theorem 3; started later, it leaves a residual term that decays with the dynamics. For other pairs the sign of $\kappa - 1$ is not determined, so $\kappa \leq 1$ is a restriction of this paper’s model; the POMDP sweeps κ over $[0, 1]$, independently of ρ .

4. Structure of the optimal policy

This section states what can be proved for every parameter value; the statements of Lemma 1 and Propositions 1–3, and all proofs, with the status of each, are in Section S1 of the Supplementary Material. Lemma 1, Proposition 1, Theorem 1 and Proposition 2 are classical results specialised to this model and are not claimed as new (Smallwood and Sondik, 1973; Blackwell, 1953; Lovejoy, 1987; Krishnamurthy, 2025); they are stated so that the assumptions each needs here are explicit, and the numerical study uses them as checks of the solver.

4.1. Concavity

By Lemma 1, a throttled observation is a garbling, in the sense of Blackwell (1953), of a monitored one: the Gaussian law at separation κd is the law

at separation d passed through a noise kernel common to both causes, and so it is worth less to an operator whose continuation value is concave. By Proposition 1, the optimal value and every Q -function are concave and continuous in the belief, in the two-state and the three-state model; under the Bayes-consistent quadrature rule so are the grid values, which the numerical study uses as a test of the solver.

4.2. Monotonicity in discriminability and the critical factor

Theorem 1 (Monotonicity in discriminability; a single critical factor). Under Definitions 1–7, 10 and Assumptions 2–4, with all other parameters fixed: (i) for $0 \leq \kappa < \kappa' \leq 1$, $V_t(\cdot, v; \kappa) \geq V_t(\cdot, v; \kappa')$ for every t and v , in the two-state and in the three-state model; (ii) in the two-state model, for $0 \leq d < d'$, $V_t(\cdot, v; d) \geq V_t(\cdot, v; d')$ for every t and v (the monitoring separation is d and the throttling separation κd in both); (iii) at every prior p_0 , the value of throttling $G(\kappa)$ is non-negative and non-decreasing on $[0, 1]$, so $\{\kappa \in [0, 1] : G(\kappa) > 0\}$ is empty, $(\kappa^*, 1]$ or $[\kappa^*, 1]$; (iv) for $\kappa < \kappa^*$, $G(\kappa) = 0$: an optimal policy of the recursion without throttle is optimal from p_0 , so an optimal operator need never throttle; for $\kappa > \kappa^*$, $G(\kappa) > 0$ and every policy that is optimal from p_0 throttles with positive probability. (iii) and (iv) hold in the three-state model with p_0 replaced by π_0 .

Theorem 1(iii)–(iv) makes the critical factor κ^* (Definition 10) a single threshold: below it the throttle has no value and a policy that never throttles is optimal; above it, every optimal policy throttles with positive probability. Theorem 1 does not locate κ^* : Proposition 6 bounds it away from zero, Proposition 5 locates it in a diffusion model, and whether $\kappa^* < 1$ on the calibrated region is claim 3, tested numerically.

4.3. Kill and leave regions

Proposition 2 (stated and proved in Section S1 of the Supplementary Material) gives the classical shape of the stopping regions. In the two-state model the optimal value is C_K at $p = 1$ and H_F at $p = 0$; kill is optimal on a closed interval $[p_t^K(v), 1]$ and leave on a closed interval $[0, p_t^L(v)]$, with $p_t^L(v) \leq q^* \leq p_t^K(v)$ for the kill/leave indifference belief q^* of Assumption 1. In the three-state model both sets are convex, the kill set containing the vertex e_K and the leave set the edge $[e_F, e_B]$. Between the two regions lie the beliefs at which it pays to keep watching, to throttle or to verify; Proposition 2 does not say how these are arranged, and part (d) of Section 5.6 tests whether no monitored belief lies above a throttled one.

4.4. Embedding of the one-shot rule

Proposition 3 (stated and proved in Section S1 of the Supplementary Material) embeds Paper 2's one-shot rule. With Paper 2's delay harm read as $\ell_v r_K$ and $T \geq \ell_v$, the rule P2 of Definition 9(iv) has expected cost $J^{P2} = \min\{R, R_v\}$, the smaller of Paper 2's passive and verify-first risks. When order flow is uninformative ($d = 0$) the optimal value equals it, $V_0(p, 0) = J^{P2}(p)$, and with positive harm and throttle cost the optimal policy is P2 up to the tie-breaking at one boundary; for every $d \geq 0$, $V_0(p, 0; d) \leq J^{P2}(p)$ (part (iv)).

That the sequential optimum is strictly better when order flow is informative is claim 2, a corollary of Proposition 3 whose magnitude the numerical study reports.

4.5. A blinding throttle is never used

Proposition 6 (A blinding throttle is never used). Under Definitions 1–7, 9, 10, Assumptions 1–3 and Proposition 1 (continuity; used

in (iii) for the value V^{KSV} of a restricted recursion of Definition 9), in the two-state and in the three-state model, with all parameters but κ fixed: (i) $V_t(\cdot, v) \leq V_{t+1}(\cdot, v)$ for $0 \leq t < T$ and $v \in \{0, 1\}$, and the same holds for the values V_t^{KS} , V_t^{KSV} , V_t^{KST} of the restricted recursions of Definition 9; (ii) at $\kappa = 0$, $Q_t(p, v, \text{throttle}) \geq V_t(p, v) + \delta + \rho p r_K$ for every $t < T$, p and v (in the three-state model with $\pi(K)$ in place of p), and $V_t(\cdot, v; 0) = V_t^{\text{KSV}}(\cdot, v)$; in particular $G(0) = 0$ at every prior; (iii) if $\delta > 0$, there is $\kappa_0 \in (0, 1]$, which does not depend on the prior, such that for every $\kappa \in [0, \kappa_0]$ **throttle** is not optimal at any decision state and $G(\kappa) = 0$; hence $\kappa^* \geq \kappa_0 > 0$.

A throttle with $\kappa = 0$ is a paid pause: it leaves the belief where it was, costs the disruption and the reduced harm of the period, and shortens the time left to decide, which by part (i) can only raise the cost to go. So $G(0) = 0$ at every prior, and with a disruption cost part (iii) extends this to an interval of factors, so $\kappa^* > 0$. Part (iii) uses the continuity of the restricted value V^{KSV} ; the proof obtains it through part (ii), by which $V^{\text{KSV}} = V(\cdot; 0)$, a value to which Proposition 1 applies.

4.6. A cost-per-information law in a diffusion model

In a continuous-time version of the two-state model, without a horizon, discounting or **verify**, the location of κ^* has a closed form.

Definition 12 (Diffusion model). The two-state model in continuous time $t \geq 0$, with no horizon, no discounting and no **verify**. A *policy* is given on a filtered probability space carrying M (with $P(M = K) = p_0$) and a standard Brownian motion B independent of M (weak formulation): a posture process $x_t \in \{\text{monitor}, \text{throttle}\}$ and a stopping time $\tau < \infty$ a.s., both with respect to the filtration $(\mathcal{F}_t^Y)$ of the order flow

$$Y_t = \int_0^t \kappa_{x_s} d\theta_M ds + B_t, \quad \theta_K = +\frac{1}{2}, \theta_F = -\frac{1}{2}, \quad \kappa_{\text{monitor}} = 1, \kappa_{\text{throttle}} = \kappa \in [0, 1],$$

with x progressively measurable; at τ the better terminal action at p_τ is taken, where $p_t = P(M = K \mid \mathcal{F}_t^Y)$. Its expected cost is $J(p_0) = \mathbb{E}[\int_0^\tau c_{x_t}(p_t) dt + g(p_\tau)]$, with running costs $c_{\text{monitor}}(p) = p r_K$, $c_{\text{throttle}}(p) = \delta + \rho p r_K$ and $g(p) = \min\{p C_K + (1-p) C_F, p H_K + (1-p) H_F\}$. $V^D(p_0)$ is the infimum of J over policies and $V^{D,\text{KS}}(p_0)$ the infimum over policies with $x \equiv \text{monitor}$; $G^D(\kappa) = V^{D,\text{KS}} - V^D$ and $\kappa_D^* = \inf\{\kappa \in [0, 1] : G^D(\kappa) > 0\}$ ($\kappa_D^* = 1$ if the set is empty). The *throttle-free continuation interval* is $\mathcal{C}_0 = \{p : V^{D,\text{KS}}(p) < g(p)\}$, with $\bar{p}_0 = \sup \mathcal{C}_0$; $\mathcal{C}_1$ (with $\bar{p}_1 = \sup \mathcal{C}_1$) is the set $\{V^D < g\}$ of the model with $\kappa = 1$. The *observability-blind policy* NB^D applies to the true belief the posture and stopping rule of Proposition 5(ii) computed with $\kappa = 1$; $J^{\text{NB}}(p_0; \kappa)$ is its expected cost in the model with the true κ .

Proposition 5 (Index law in the diffusion model). Under Definition 12 and Assumptions 1 and 4, with $d > 0$ and $r_K > 0$, let $c^*(p) = \min\{p r_K, (\delta + \rho p r_K)/\kappa^2\}$ ($c^*(p) = p r_K$ if $\kappa = 0$), and let $\tilde{P}$ be the belief of the monitoring model, $d\tilde{P}_u = d\tilde{P}_u(1 - \tilde{P}_u) dW_u$, $\tilde{P}_0 = p$, with W a standard Brownian motion. (i) (*Reduction to optimal stopping.*) $V^D(p) = \inf_\sigma \mathbb{E}_p[\int_0^\sigma c^*(\tilde{P}_u) du + g(\tilde{P}_\sigma)]$, the infimum over a.s. finite stopping times of $\tilde{P}$. (ii) (*Index.*) If $\delta > 0$, $\rho > 0$ or $\kappa = 0$, there is an optimal policy that stops when p_t leaves the continuation set of (i) and, before stopping, throttles at belief p if and only if $\kappa^2 > \rho + \delta/(p r_K)$, and monitors otherwise. When $\kappa^2 > \rho$, the throttled beliefs are those above $p_{\text{thr}} = \delta/((\kappa^2 - \rho) r_K)$. If $\delta = \rho = 0 < \kappa$, $V^D(p) = p C_K + (1-p) H_F$ and, for $p \in (0, 1)$, no policy is optimal. (iii) (*Critical factor.*) $\mathcal{C}_0 = (\underline{p}_0, \bar{p}_0)$ is an open interval that contains q^* . For $p_0 \in \mathcal{C}_0$, $G^D(\kappa) > 0$ if and only if $\kappa^2 > \rho + \delta/(\bar{p}_0 r_K)$; hence $\kappa_D^* = \min\{1, \sqrt{\rho + \delta/(\bar{p}_0 r_K)}\}$, and $\kappa_D^* = \sqrt{\rho}$ when $\delta = 0$, whatever the loss table, d and $p_0 \in \mathcal{C}_0$. In particular, if $\kappa = r$ and $\rho = r^\alpha$ for some $r \in (0, 1)$

and $\alpha > 0$ (a scaling throttle, Proposition 4(i), with harm of order α in the injected flow), then $G^D(\kappa) > 0$ requires $\alpha > 2$. (iv) (*Observability-blind operator*.) Suppose $\rho < 1$, $\delta > 0$ or $\rho > 0$, $p_{\text{NB}} = \delta/((1 - \rho)r_K) < \bar{p}_1$ and $p_0 \in \mathcal{C}_1$. Then $\mathcal{C}_1 = (\underline{p}_1, \bar{p}_1)$ is an open interval containing q^* ; $J^{\text{NB}}(p_0; \kappa)$ is continuous and strictly decreasing in $\kappa \in (0, 1]$, tends to $+\infty$ as $\kappa \downarrow 0$, and equals $J(p_0)$ for the solution J of $\frac{1}{2}d^2p^2(1 - p)^2J''(p) = -\hat{c}_\kappa(p)$ on $\mathcal{C}_1$ with $J = g$ at $\underline{p}_1$ and $\bar{p}_1$, where $\hat{c}_\kappa(p) = pr_K$ for $p \leq p_{\text{NB}}$ and $(\delta + \rho pr_K)/\kappa^2$ for $p > p_{\text{NB}}$. Hence $\{\kappa \in (0, 1] : J^{\text{NB}}(p_0; \kappa) > V^{D, \text{KS}}(p_0)\} = (0, \kappa_{\text{NB}})$ for a unique $\kappa_{\text{NB}} \in (0, 1]$.

Proposition 5 is claim 7(a); its full proof is in Section S1 of the Supplementary Material. A throttled period yields information at the rate κ^2 relative to monitoring, since the Kullback–Leibler divergence between two Gaussian laws is quadratic in the distance between their means. Part (ii) says that, when $\delta > 0$ or $\rho > 0$, the throttle is worth using at a belief exactly when its running cost per unit of information, $(\delta + \rho pr_K)/\kappa^2$, is below monitoring’s, pr_K . “Index law” is shorthand for this comparison, not a Gittins or bandit index. In the excluded case $\delta = \rho = 0 < \kappa$ the throttle is free and informative, and no policy attains the infimum. Because harm accrues only under a compromise, the operator monitors at low beliefs and throttles at high ones. Part (iii) gives κ_D^* in closed form, and, since a scaling throttle has $\kappa = r$ (Proposition 4(i)), such a throttle is worth using only if harm falls faster than quadratically in the throttled flow. Part (iv) makes the blind operator of claim 4 exact: its cost grows without bound as κ falls, so it does worse than the best kill-switch for every κ below some $\kappa_{\text{NB}} > 0$. The proof (Section S1) uses the time change to the information clock of Campbell et al. (2025); relative to that work, the running cost depends on the belief, so the optimal posture is a switch rather than a constant, and the action acts on

harm and information at once.

Model D is a limit of the discrete model (period length to zero at fixed separation per unit time), which motivates it; this is not a result of the paper. The discrete model has a deadline, so its κ^* need not equal κ_D^* ; whether the law locates it on the calibrated region is claim 7(b), and it does not (Section 5.6).

4.7. *What the results do not give*

Weak dominance, $\Gamma(p_0) \geq 0$, holds by construction and is not a finding. For the discrete model the results do not say when the gain is strict, where κ^* lies in $(0, 1]$, or how much a blind operator loses; these are the questions of Section 5. In the three-state model, Lemma 1, Proposition 1(ii), Theorem 1(i), (iii)–(iv) and Proposition 2(iv) hold on the 2-simplex. Verification reads authorization, which is valid under both a fault and a benign shock: under Definition 4 with $t_B = t_F$, its outcome has the same law under F and B , so only order flow separates them.

5. Numerical study

This section tests the paper’s substantive claims by an exact computation. The design, the tolerance, the materiality margin, the calibrated region and the criteria that would support, weaken or refute each claim were fixed before any result existed, in a written experiment brief (B001, 29 September 2026; four amendments, all before any result, none of which changes a prediction or a criterion). Section S2 of the Supplementary Material describes the amendments and lists the commit and merge time of the brief and of each amendment beside the first output of the computation; the brief was not lodged with an external registry.

5.1. Questions

Of the paper’s seven claims, five are computed: (1) is $\Gamma(p_0)$ strictly positive and material, and how much of it is the throttle’s; (2) how large is the gap to Paper 2’s one-shot rule when $d > 0$ (a corollary of Proposition 3); (3) is $\kappa^* < 1$ on the calibrated region; (4) for which κ does NB cost more than KS*; (5) does a benign third cause change the policy materially. Claim 6 is Proposition 4 and claim 7(a) is Proposition 5. Claim 7(b) is tested by a second brief (B002), which fixed its own tolerances and bands before any result.

5.2. Calibrated region

The region is built from Paper 1’s cost study (Mejrissi, 2026b) and a box, fixed in the brief, for what Paper 1 does not determine (Table S2 and Section S3 of the Supplementary Material). Its cells are the set $\mathcal{F}$ of Paper 1’s cells without a supported improvement over the best constant policy; each supplies its loss table, with the measured cost of leaving a fault, and its prior p_0 . The equivalent-Gaussian calibration (Remark 1) fixes the per-period separation $d = d_W / \sqrt{n_W}$, with n_W a modelling choice. The box crosses the window length, the harm and disruption scales, ρ , the verification accuracy and a sweep of κ over $[0, 1]$, so that each cell contributes 2268 points. The other cells of Paper 1’s table and a direct sweep of d are computed as a contrast set, outside the region.

5.3. Metrics and decision criteria

Each point is measured against the *residual ambiguity* $\Xi(p_0) = V_0^{\text{KS}}(p_0, 0) - O(p_0)$, what the best kill-switch still loses to an operator who knows the cause, whose cost is $O(p_0) = p_0 C_K + (1 - p_0) H_F$. A point is *grid-converged*

(Definition 8, Supplementary Material) if doubling N and n_q changes every reported value by at most $\epsilon = 10^{-3} \Xi(p_0)$. The graded gain is *material* if $\Gamma \geq m \Xi(p_0)$ with $m = 0.05$, and the gap to P2 if it recovers the same share of what P2 loses to the oracle. κ^* is located by bisection, and κ_{NB} is the largest κ of the sweep at which J^{NB} exceeds V_0^{KS} beyond the tolerance. Claim 1 is refuted if the graded gain is zero up to the tolerance at every point, and weakened if it is material at no more than half of the points, comes entirely from verification, or is material only outside the region. Claim 2 is refuted if the gap is zero up to the tolerance at every point with $d > 0$, and weakened if it is material at no more than half of them. Claim 3 is refuted if $\kappa^* = 1$ at every point of the region, and weakened if $\kappa^* < 1$ only outside it. Claim 4 is refuted if NB never costs more than KS* beyond the tolerance, and weakened if it does so only outside the region. A claim is *inconclusive* if more than a quarter of its points are not grid-converged after two doublings or straddle the materiality threshold. Claim 5’s criteria are given with its evidence (Section S3).

5.4. Checks and convergence

Concavity (Proposition 1(iii)), monotonicity of G in κ (Theorem 1), equality with Paper 2’s rule at $d = 0$ (Proposition 3(ii)), a zero value of verification on the F – B edge and the absence of $\kappa^* = 0$ (Proposition 6) were checked before the sweep was read; none failed (Section S4). Of the 22680 points of the region, 20256 are grid-converged after two doublings; the unconverged points and the converged points that straddle the materiality line together are a share 0.123 of the points, below the quarter that would make claim 1 inconclusive. The sweep used 10.4 CPU-hours.

Table 1: Outcome of claims 1–5 on the calibrated region, by the criteria fixed in brief B001 before any result. Row: the brief’s label of the claim.

Row	Claim	Outcome	Deciding figures
1	1	supports	material at 16125 of 20256 converged points (0.796); lowest cell 0.556
1b	2	weakens	material at 3707 of 20256 converged points (0.183)
2a	3	supports	κ^* interior at 10, = 1 at 910, = 0 at 0 of 920 converged groups
2b	4	supports	κ_{NB} defined in 8 groups
4	5	inconclusive	227 of 600 points unusable, 16 usable points straddling (share 0.405)

5.5. Results

Table 1 gives the outcome of claims 1–5; Section S4 of the Supplementary Material gives the figures per cell, the three-state set and the optimal-action maps. Negative outcomes are reported beside positive ones.

Claim 1: supported. The graded gain is material at 16125 of the 20256 converged points (share 0.796), and at more than half of the converged points in each of the 10 cells of $\mathcal{F}$. It does not come entirely from verification, but the throttle share exceeds the grid tolerance at only 644 of the material points: the gain is not a throttling result. The operating characteristics of each policy are not tabulated in this version.

Claim 2: weakened. The gap to Paper 2’s one-shot rule is positive beyond the tolerance at 4681 points but material at only 3707 of the 20256 converged points (share 0.183), no more than half: the value of sequential evidence is real but, on most of the region, below the materiality margin.

Claim 3: supported, on few groups. Among the 920 converged κ -groups of the region, $\kappa^* = 1$ in 910 and $\kappa^* < 1$ in 10, between 0.219 and 0.769, from few cells. The claim is supported, since $\kappa^* < 1$ on a non-empty part of the region, but elsewhere the throttle has no value at any factor below one. The predicted increase of κ^* with δ and ρ holds only as non-decreasing: strict increases are rare, most ties being at $\kappa^* = 1$ (Section S4).

Claim 4: supported, on the same groups. The blind policy costs more than the best kill-switch beyond the tolerance, for κ below a κ_{NB} , in 8 κ -groups of the region, 8 of them among those with $\kappa^* < 1$: the region where ignoring the observability price is costly is non-empty, and as narrow as the region where the throttle has value at all.

Claim 5: inconclusive at this power. Too many three-state points are not grid-converged or straddle the materiality line (share 0.405, above the quarter fixed in advance), so nothing is read from the usable points. The design, criteria and figures are in Sections S3 and S4 of the Supplementary Material.

5.6. Does the diffusion law carry over? (Claim 7(b))

At each converged point the second brief forms, from the upper edge $\bar{p}_0$ of the throttle-free continuation set, the law's prediction $\hat{\kappa}^* = \min\{1, \sqrt{\rho + \delta/(\bar{p}_0 r_K)}\}$ (Proposition 5(iii)). It has four parts: (a) *location*, the share of κ -groups with $\hat{\kappa}^* < 1$ at which κ^* lies within a tolerance of $\hat{\kappa}^*$, which supports claim 7(b) above three quarters, weakens it between a quarter and three quarters and refutes it at a quarter or less; (b1) whether a scaling throttle with linear harm has no value; (b2) on a slice with $\delta = 0$, whether κ^* lies within the tolerance of $\sqrt{\rho}$; (c) whether the closed-form policy IDX, which throttles if

Table 2: Outcome of each part of the test of claim 7(b), by the criteria fixed in brief B002 before any result. n : usable κ -groups (parts a, b2) or points (b1, c, d); Share: the share the criterion is read on. For part d, “supports” means that the band order holds.

Part	Outcome	n	Share
a	refutes	690	0.057
b1	supports	636	0.000
b2	weakens	18	0.556
c	supports	5905	0.994
d	supports	920	0.880

and only if $\kappa^2 > \rho + \delta/(pr_K)$ where the throttle-free optimal action is **mon-**
itor, has regret at most a fifth of the graded gain; (d) whether no monitored
belief lies above a throttled one. Claim 7(b) is supported if (a) and (c) fall in
their support bands, weakened if either falls in its weaken band, and refuted
if (a) falls in its refute band (Table 2).

Claim 7(b): refuted. The law does not locate the discrete κ^* : among the 690
converged κ -groups with $\hat{\kappa}^* < 1$, κ^* lies within the tolerance at 39 (share
0.057), in the refute band. The miss is one-sided: where the law predicts that
throttling has value, the discrete model almost never throttles, and where it
predicts none (230 groups) the discrete model throttles in 0. (b1) supports:
a scaling throttle with linear harm has no value, as the law predicts. (b2)
weakens: with no disruption cost, κ^* lies within the tolerance of $\sqrt{\rho}$ in a
share 0.556 of the groups. (c) supports: IDX’s regret is at most a fifth of
the gain at a share 0.994 of the points where the gain is material; this is not
evidence that the law locates κ^* , since the throttle carries little of the gain
(claim 1). (d) holds at a share 0.880 of the points where the optimal policy
throttles. Claim 7(a), about the diffusion model, is unaffected.

6. Discussion

What is claimed, and at what strength. The observability price has two forms. For an optimal operator it is a threshold: below κ^* the throttle has no value (Theorem 1), which says when to throttle, not that throttling is harmful. For an operator who plans as if throttling kept full discriminability it is a loss, unbounded as κ falls in the diffusion model and, in the discrete model, worse than the best kill-switch on part of the region. On the calibrated region the observability price is shown to exist, not to be common: the throttle carries little of the graded gain, and $\kappa^* < 1$ and the blind operator’s loss occur on few groups. One possible reason for the one-sided miss of claim 7(b), a hypothesis this paper does not test, is that verification and the finite horizon, both absent from the diffusion model, supply the information that the throttle would buy.

Modelling choices. The POMDP treats ρ and κ as independent and κ as constant, which in Paper 1’s linear model is exact only for a throttle in force from the onset (Proposition 4). Throttles that make the causes easier to tell apart, for instance because a persistent adversary pushes against the limit, are excluded by Assumption 4. The separation is backed out of a classifier accuracy through a Gaussian benchmark that Paper 1’s simulated constructions were never shown to satisfy, and it is a separation for a whole window; it is therefore labelled a heuristic, converted to a per-period value under a stated window length, and accompanied by a direct sweep of d . The operator knows every parameter and the cause does not react (Assumption 5); the strategic case is the subject of Paper 4 (Mejrissi, 2026a). In the three-state extension verification separates a compromise from the other two causes but not a fault from a benign shock; whether this changes the policy materially

is claim 5, inconclusive at this power, so the extension is a method, not yet a result.

7. Conclusion

Responding to an ambiguous incident on a trading gateway is a sequential decision in which containment has a price in evidence. Classical results, specialised to the model, give a single critical factor κ^* below which an optimal operator need never throttle, and recover Paper 2’s one-shot verification rule when order flow is uninformative. Two results are specific to graded containment: a throttle that blinds too much is never used, so $\kappa^* > 0$ with a disruption cost; and, in a diffusion model with a disruption cost or residual harm, the throttle is worth using exactly where its cost per unit of information is below monitoring’s, which gives κ^* in closed form and makes the loss of an operator blind to the observability price unbounded as κ falls. An exact computation with criteria fixed in advance shows that, on a region calibrated from Paper 1’s cells, the graded policy strictly and materially beats the best kill-switch at most points, with little of the gain from the throttle; that $\kappa^* < 1$, and the blind operator does worse than the kill-switch, only on a small part of the region; that the gain over the one-shot rule is weakened and the effect of a benign third cause inconclusive at this power; and that the diffusion law does not locate the discrete κ^* , although its predictions for scaling throttles and for the band order hold; its prediction $\kappa^* = \sqrt{\rho}$ without disruption cost is weakened.

Data and code availability

The solver that computes the optimal policies and the reference policies of Section 5, the experiment configurations, the provenance register and the result files from which every number in this article is generated are held in the author’s project repository. They will be deposited in a public archive on acceptance, with a persistent identifier (DOI to be determined). The briefs that fix the design and the decision criteria of Section 5 (B001, file `papers/paper3/research/briefs/B001_graded-vs-killswitch-solver.md`, and B002) and their amendments are held in the same repository, whose commit history timestamps each of them, and will be deposited with it; until then they are available from the corresponding author on reasonable request. The loss tables, priors and classification accuracies that calibrate the numerical study are those of the companion study (Mejrissi, 2026b), whose deposit they share.

CRedit authorship contribution statement

Chiheb Mejrissi: Conceptualization, Methodology, Formal analysis, Software, Investigation, Validation, Visualization, Writing – original draft, Writing – review & editing.

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References

- Blackwell, D., 1953. Equivalent comparisons of experiments. *The Annals of Mathematical Statistics* 24, 265–272. doi:10.1214/aoms/1177729032.
- Campbell, S., Gaitsgori, G., Groenewald, R., 2025. A sequential testing problem with signal control. *arXiv preprint 2509.18209*. URL: <https://arxiv.org/abs/2509.18209>, arXiv:2509.18209.
- Chernoff, H., 1959. Sequential design of experiments. *The Annals of Mathematical Statistics* 30, 755–770. doi:10.1214/aoms/1177706205.
- Hammar, K., Dhir, N., Stadler, R., 2026. Optimal defender strategies for CAGE-2 using causal modeling and tree search. *IEEE Transactions on Network and Service Management* doi:10.1109/TNSM.2026.3735865. early access.

- Hammar, K., Stadler, R., 2022. Intrusion prevention through optimal stopping. *IEEE Transactions on Network and Service Management* 19, 2333–2348. doi:10.1109/TNSM.2022.3176781.
- Harrison, J.M., Sunar, N., 2015. Investment timing with incomplete information and multiple means of learning. *Operations Research* 63, 442–457. doi:10.1287/opre.2015.1344.
- Kazeminajafabadi, A., Imani, M., 2024. Optimal joint defense and monitoring for networks security under uncertainty: A POMDP-based approach. *IET Information Security* 2024, 7966713. doi:10.1049/2024/7966713.
- Krishnamurthy, V., 2025. Partially Observed Markov Decision Processes. 2 ed., Cambridge University Press. doi:10.1017/9781009449441.
- Lovejoy, W.S., 1987. Some monotonicity results for partially observed Markov decision processes. *Operations Research* 35, 736–743. doi:10.1287/opre.35.5.736.
- Mejrissi, C., 2026a. Attack–fault ambiguity against a strategic attacker: The stealth–harm frontier, exploitable commitments, and the verification margin. Working paper.
- Mejrissi, C., 2026b. Attack–fault confounding at the electronic trading gateway is fragile. Working paper.
- Mejrissi, C., 2026c. Verification as a costed action: When checking beats containing under attack–fault ambiguity. Working paper.
- Miehling, E., Rasouli, M., Teneketzis, D., 2018. A POMDP approach to the dynamic defense of large-scale cyber networks. *IEEE Transactions on*

Information Forensics and Security 13, 2490–2505. doi:10.1109/TIFS.2018.2819967.

Naghshvar, M., Javidi, T., 2013. Active sequential hypothesis testing. The Annals of Statistics 41, 2703–2738. doi:10.1214/13-AOS1144.

Naha, A., Teixeira, A., Ahlén, A., Dey, S., 2023. Quickest physical watermarking-based detection of measurement replacement attacks in networked control systems. European Journal of Control 71, 100804. doi:10.1016/j.ejcon.2023.100804.

Nitinawarat, S., Atia, G.K., Veeravalli, V.V., 2013. Controlled sensing for multihypothesis testing. IEEE Transactions on Automatic Control 58, 2451–2464. doi:10.1109/TAC.2013.2261188.

Smallwood, R.D., Sondik, E.J., 1973. The optimal control of partially observable Markov processes over a finite horizon. Operations Research 21, 1071–1088. doi:10.1287/opre.21.5.1071.